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Multilingual Natural Language Processing

Mini Project

Practical Mini Project

- ▶ Work on **one** concrete multilingual NLP problem
- ▶ Canvas course web page lists example problems and links to relevant data sets
 - but you can also choose your own problem
- ▶ Work in teams of 5 students

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- ▶ In week 7 each team gives a 10-15 minutes presentation followed by 5 minutes for questions

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- ▶ Deadline for submitting presentations/reports/code is Friday, October 9, 12:00 noon

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 - choose your team
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- ▶ Week 2:
 - identify and read relevant previous literature
 - inspect data, pre-process, test/valid data
 - additional data crawling?
 - what should be the initial model?

Practical Mini Project

► Week 3:

- implement first model
- evaluate
- debug?
- what works, what doesn't?

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▶ Week 4:

- refine first model/try alternative model
- bells and whistles (dropout, layernorms, more layers, residual, highway connections, etc.)
- evaluate and compare
- what works, what doesn't?

Practical Mini Project

► Week 5:

- refine second model/try (another) alternative model
- error analysis, open problems
- conclusions, recommendations

Practical Mini Project

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 - refine second model/try (another) alternative model
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- ▶ Week 6:
 - address any loose ends
 - write 4 page report with main findings
 - prepare slides for presentation

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 - as link to Github repo (preferably)
or gzipped tarball
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- ▶ Team presentation slots (for week 7) will be assigned at the beginning of week 6

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- ▶ Some practical advice
 - keep the data small, most likely, you'll be running it on your laptop
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 - thorough evaluation (scores under different settings/architectures); error analysis

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 - good presentation focusing on the most important aspects and finding, understandable by wider audience, contains examples

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 - anything that contradicts previous literature (bug or finding)?